FOLINIC ACID

Methotrexate rescue

Never intrathecal

CHEMICAL NAME (2S)-2-[[4-[(2-amino-5-formyl-4-oxo-1,4,5,6,7,8-hexahydropteridin-6-yl)methylamino]benzoyl]amino]pentanedioic acid (calcium folinate)

CLASS Reduced folate (rescue agent and 5-FU modulator)

TRADE NAMES Leucovorin calcium · Calcium folinate · Levofolinate (Isovorin)

THERAPEUTIC USES / INDICATIONS

- Rescue after high-dose methotrexate (and methotrexate overdose/toxicity).
- Enhances 5-fluorouracil in colorectal cancer (FOLFOX, FOLFIRI).
- Megaloblastic anaemia due to folate deficiency when oral folic acid unsuitable (rare).
- Toxicity from trimethoprim, pyrimethamine.

ADMINISTRATION

- **IV** (slowly — max 160 mg/min because of calcium content), **IM** or **PO**.
- Methotrexate rescue: start at a fixed time (usually 24–36 h after MTX start) and continue until MTX level is safe.
- Never give intrathecally (fatal after intrathecal methotrexate overdose).

DOSAGES

- **High-dose MTX rescue:** typically 15 mg (10–15 mg/m²) every 6 h for 10–12 doses, adjusted to MTX levels.
- **With 5-FU (FOLFOX):** 400 mg/m² (racemic) or 200 mg/m² (levofolinate) IV over 2 h.
- **MTX overdose:** dose equal to or greater than MTX dose, as soon as possible.

PHARMACOKINETICS

- **Onset:** IV immediate; PO 30 min (saturable absorption >25 mg).
- Converted to active **5-methyltetrahydrofolate** (does not need DHFR).
- Excreted in urine. **t½:** ~6 h (active metabolites).

PHARMACODYNAMICS

- Provides **reduced folate** that bypasses DHFR blockade → rescues normal cells from methotrexate toxicity.
- Stabilises 5-FU (FdUMP)–thymidylate synthase complex → ↑ 5-FU cytotoxicity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Rare: allergic reactions.
- Nausea (oral).
- Fever.

ADVERSE EFFECTS (SERIOUS)

- ↑ 5-FU toxicity (diarrhoea, mucositis); intended synergy.
- Seizures (in epileptics — reduces antiepileptic levels).
- Hypersensitivity.

CONTRAINDICATIONS & CAUTIONS

- Pernicious anaemia or B12-deficiency megaloblastic anaemia (masks deficiency).
- Intrathecal administration.

KEY INTERACTIONS

- **5-FU, capecitabine:** ↑ efficacy and toxicity.
- **Phenytoin, phenobarbital, primidone:** ↓ levels → seizures.
- **Co-trimoxazole:** may ↓ efficacy for PCP treatment.

NURSING MANAGEMENT

- Give rescue doses exactly on time; delays cause severe toxicity.
- Monitor MTX levels, renal function, urine pH (alkalinisation >7), fluid balance with high-dose MTX.
- IV slowly; check route.

PATIENT EDUCATION

- Rescue doses must be taken on time, including at night.
- Report mouth sores, diarrhoea, fever.
- Drink plenty of fluids after high-dose methotrexate.

KEY POINTS

- Reduced folate — bypasses DHFR.
- Potentiates 5-FU (FOLFOX).
- Methotrexate rescue on strict schedule.
- Never intrathecal; IV ≤160 mg/min.